AIS validation — production run results

Generated by `examples/validation/run_case.jl` + `analyze.jl` on 10 threads.

What this tests

The annealed importance sampler (AIS) and a standard cycle walk are each run against the **same** target measure. If both are correct they must sample the same distribution, so we compare the distribution of **district isoperimetric scores** (`perimeter² / area`, higher = less compact) between the two runs.

For every plan we sort its district scores ascending — rank 1 = most compact district, rank K = least compact — turning each plan into order statistics. For each rank we compare the AIS marginal (**importance-weighted** by `exp(log-weight)`) against the cycle-walk marginal (unweighted): weighted mean, weighted variance, and the total variation (TV) distance between binned histograms.

There is no analytic answer, so each between-method TV is judged against a **split-half noise floor**: the TV between the two halves of each run on its own. Agreement means the AIS-vs-cyclewalk TV sits within that finite-sample noise (verdict `ok`). The AIS **effective sample size** `ESS = (Σw)²/Σw²` is reported first — if it is a small fraction of the sample count the importance weights have collapsed and the comparison is unreliable regardless of the TV.

For the **small** case only, we also compare the full distribution over concrete partitions directly (district labels canonicalized, no graph-symmetry folding), which doubles as a mixing check, again against a split-half state-TV noise floor.

Target and sizes per case

case	graph	districts	target measure	proposal mix
small	`4x4pct_2x2cnty.json`	4	`gamma=0.7` spanning forests, **no iso** (matches AIS unit test)	0.5 / 0.5
ct	`CT_pct20.json`	5	`gamma=0.25` + `iso_weight=0.3`	0.1 / 0.9
grid	`grid_graph_10_by_10.json`	3	`gamma=0.25` + `iso_weight=0.3`	0.1 / 0.9

Production sizes (both modes): AIS = 10,000 samples, 4,000 annealing steps per sample, 500 base-chain steps between samples; cycle walk = 10,000 samples, 1,000 steps apart.

Runtimes

Wall-clock, all runs on one Apple-Silicon machine with `julia -t 10` (10 threads):

case	mode	anneal steps	wall time
small	AIS	4,000	66 s

case	mode	anneal steps	wall time
small	cycle walk	—	68 s
grid	AIS	4,000	81 s
grid	AIS	16,000	269 s (4:29)
grid	cycle walk	—	102 s
ct	AIS	4,000	295 s (4:55)
ct	AIS	16,000	1157 s (19:17)
ct	cycle walk	—	441 s (7:21)

Notes:

- **Cold start.** Each row is a fresh `julia -t 10` process, so every time includes ~20–45 s of Julia startup + JIT compilation. That fixed cost dominates the sub-100 s small/grid runs (see below) and should be subtracted before reasoning about sampler throughput.
- **AIS parallelism pays off.** AIS anneals `ntasks = 10` samples concurrently, so even though it does more total MCMC work, CT AIS-4k (295 s) finished *faster* than the strictly serial CT cycle walk (441 s).
- **Annealing scales ~linearly with the schedule.** Grid went 81 s → 269 s for a 4× schedule. Fitting `fixed + anneal = 81`, `fixed + 4·anneal = 269` gives fixed overhead ≈ 18 s and a 4k-annealing phase ≈ 63 s — i.e. the annealing cost is very nearly proportional to `--anneal-steps`, as expected. CT's 16k rerun therefore lands near 4× its 4k annealing phase plus fixed overhead.
- **CT is the cost driver** because each annealing step evaluates a per-district log-determinant *twice* (the weight-tracking prestep hook reads the log-energy before and after each measure update), and CT has 5 districts on ~700 nodes. BLAS is pinned to 1 thread during multi-task AIS so the 10 annealing tasks don't oversubscribe the cores with competing BLAS pools.

Methodology: the split-half noise floor

The problem it solves

We are comparing two *finite* samples and asking "do these come from the same distribution?" There is no analytic target here, so a raw total-variation distance is uninterpretable on its own: **even two samples drawn from the exact same distribution have a nonzero TV**, because a histogram built from 10,000 correlated MCMC draws is not the true density — it wobbles from finite-sample scatter and from chain autocorrelation. So `TV(AIS, cyclewalk) = 0.065` means nothing until we know how large a TV *pure noise* alone produces at this sample size. The split-half floor measures exactly that.

The construction

For one run (say the cycle walk) and one statistic (say the rank-1 histogram):

1. Split the run's samples **in on-disk order** into a first half and a second half.
2. Build the histogram on each half separately (importance-weighted for AIS).
3. Take `TV(firsthalf, secondhalf)`.

Both halves are drawn from the *same* sampler targeting the *same* distribution, so whatever TV separates them is **by construction not a distributional difference** — it is the finite-sample + autocorrelation noise of that run. That number is the "floor": the smallest TV we could expect to see between two honest samples of one distribution. Splitting *sequentially* (rather than randomly) is deliberate — it lets consecutive, autocorrelated draws stay together, so the floor captures the chain's real correlation structure, not an over-optimistic i.i.d. estimate.

We compute the floor for **both** runs and take the larger: `noise = max(splithalf(AIS), splithalf(cyclewalk))`. The noisier of the two samplers sets a fair bar — for AIS this is usually the binding one, because its *effective* sample size (ESS) is below its nominal 10,000, so its histograms are inherently noisier.

The decision rule

A rank is `ok` when

```
TV(AIS, cyclewalk)  ≤  2 · noise  +  0.02
```

- The **2×** is because the between-method TV compares two independent finite samples, whereas each split-half floor is dominated by the variability of *one* run; comparing two noisy histograms admits roughly the sum of their fluctuations. (It is also mildly conservative in the other direction: a floor is computed on N/2 samples while the between-method TV uses the full N per side, so the raw floor slightly *over*-states the noise of a full-size comparison. The 2× absorbs both effects; it is a calibrated heuristic, not a formal α-level test.)
- The **+0.02** is a small absolute cushion so that ranks with a near-degenerate score (e.g. the small case's rank-3/4, where the floor is ~0.003) are not failed by trivial rounding-scale differences.

The same construction is reused for the **small case's state distribution**: split each run in half, tabulate the weighted frequency over concrete partitions, and take the TV between halves as the floor for the state-level comparison.

What it can and cannot catch

- **Catches:** a distributional difference larger than sampling noise — a real disagreement between the two samplers. This is exactly what flagged grid at 4,000 steps: `TV ≈ 0.065` against a floor of `≈ 0.025`, *and* the mean gap pointed the same way on every rank, which noise cannot do.
- **Does not catch:** a bias that is present *identically in both halves* of a single run does **not** inflate that run's floor (correctly — the floor measures noise, not bias). Such a bias instead shows up by pushing the *between-method* TV above the floor. In other words the floor is the yardstick, not the thing being measured; a shared, stationary bias is caught by the comparison, not by the floor.
- **Assumes stationarity within a run.** If a chain drifts (non-stationary), its two halves differ for a reason other than noise, which *inflates* the floor and makes the test too lenient. For AIS this risk is low: each output sample is an independent annealing of a fresh base-chain draw, so sequential order is just base-chain order, not a slowly-mixing trajectory. It is more of a concern for the single long cycle walk, which is why the ESS and mean/variance columns are reported alongside the TV rather than collapsing everything to one pass/fail.

Reading the floor next to ESS

The floor and the AIS ESS tell complementary stories. ESS says how much independent information the importance weights carry; the floor says how much two honest samples of this size *should* differ. A run can only be trusted when ESS is a healthy fraction of the sample count **and** the between-method TV sits within the floor. When ESS collapses (e.g. CT at 4,000 steps, ESS ≈ 13), the AIS histogram is effectively built from a handful of samples, its split-half floor balloons, and any "pass" is vacuous — which is why ESS is reported first and treated as a gate on the TV verdict.

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small — PASS

AIS ESS = **9698.6 / 10000** (weights spanning only 0.86–2.20 nats: the γ=0 base and γ=0.7 target are close, so the importance weights barely vary). Cycle walk: 10001 samples.

Rank marginals of the isoperimetric score (AIS / cycle walk):

rank	mean AIS / CW	var AIS / CW	TV	noise floor	verdict
1 (most compact)	20.543 / 20.591	20.248 / 20.242	0.0053	0.0150	ok
2	22.725 / 22.847	15.300 / 14.744	0.0135	0.0148	ok
3	24.672 / 24.726	2.841 / 2.387	0.0060	0.0028	ok
4 (least compact)	24.672 / 24.726	2.841 / 2.387	0.0060	0.0028	ok

Every rank's between-method TV is at or below its split-half noise floor; means agree to ≤0.12 and variances closely.

Direct partition distribution (the mixing check): both chains visited all **117** distinct partitions, with matching top-state masses (0.036 vs 0.030, …). State TV = **0.0545** vs a split-half noise floor of **0.0959** → ok. Both samplers mix fully and agree on the distribution over concrete plans.

> The earlier dev-size freeze of the cycle walk was caused by the iso penalty; with the unit-test parameters (γ=0.7, no iso) both chains mix freely.

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grid — small systematic discrepancy (likely annealing bias)

AIS ESS = **6457.8 / 10000** (65%; weights span 39.6–44.6 nats). Cycle walk: 10001 samples. Both chains mix well, but AIS scores run **consistently higher (less compact) with higher variance** than the cycle walk at every rank:

rank	mean AIS / CW	var AIS / CW	TV	noise floor	verdict
1 (most compact)	19.969 / 19.618	5.022 / 4.705	0.0682	0.0325	ok
2	22.753 / 22.311	7.258 / 6.978	0.0648	0.0236	ok
3 (least compact)	27.287 / 26.747	11.683 / 9.943	0.0647	0.0218	**DIFFERS**

The between-method TV (~0.065 at every rank) is ~2–3× the split-half noise floor (~0.025), and the mean shift is in the **same direction on all three ranks** (+0.35 / +0.44 / +0.54), so this is not Monte-Carlo noise.

Interpretation — insufficient-annealing bias, not a bug. The base measure (γ=0, no iso) prefers *less* compact plans; the target (γ=0.25 + iso=0.3) prefers *more* compact ones. If the 4,000-step inner chain does not fully equilibrate as the measure ramps, AIS samples stay biased toward the base — exactly the observed direction (higher iso, higher spread). ESS being healthy is consistent with this: the importance weights are well-behaved; it is the inner-chain mixing at each annealing temperature that is the limit.

Confirmation: `--anneal-steps 16000` (4× the schedule)

Rerunning grid AIS with a 4× longer annealing schedule (cycle walk unchanged) closes the gap, confirming the bias diagnosis:

rank	mean AIS / CW	var AIS / CW	TV	noise floor	verdict
1	19.743 / 19.618	4.853 / 4.705	0.0254	0.0140	ok
2	22.507 / 22.311	7.139 / 6.978	0.0293	0.0287	ok
3	26.919 / 26.747	10.444 / 9.943	0.0239	0.0232	ok

The mean shifts dropped by ~½–⅔ (rank-1 +0.351 → +0.125, rank-3 +0.540 → +0.172), all TVs fell to ~0.025 (at the noise floor), ESS rose **6458 → 9027** (65% → 90%), and every rank now **PASSES**. The 4,000-step result was under-annealed; the sampler is correct and converges to the cycle walk as the schedule lengthens.

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ct — ESS collapse at 4,000 steps (vacuous "pass")

AIS ESS = **12.6 / 10000** — a near-total collapse: the log weights span **27 nats** (283.5–310.5), so effectively ~13 of the 10,000 samples carry all the weight. Runtime ~5 min (AIS) + ~7 min (cycle walk).

rank	mean AIS / CW	var AIS / CW	TV	noise floor	verdict
1 (most compact)	30.412 / 27.317	19.387 / 6.488	0.5241	0.5846	ok*
2	35.899 / 30.364	18.116 / 7.111	0.6967	0.5546	ok*
3	39.948 / 33.269	19.611 / 10.134	0.7135	0.6567	ok*
4	48.763 / 36.503	44.536 / 15.584	0.7739	0.7189	ok*
5 (least compact)	54.211 / 41.795	58.067 / 35.107	0.6568	0.5764	ok*

***The "PASS" here is vacuous** — exactly the failure mode the noise-floor methodology warns about. With ESS ≈ 13 the AIS histogram is built from a handful of effective samples, so its split-half floor inflates to ~0.5–0.7 and swallows even TVs near 0.77. The substance says the two are *not* matching: AIS means are +3 to +12 higher (less compact) than the cycle walk on every rank, and AIS variances are 3–4× larger — the same under-annealing direction seen on grid, but far more severe because the γ=0 base and the γ=0.25+iso target are much further apart on the 5-district CT map. **The ESS gate fails, so this result is not trustworthy regardless of the TV verdict.** Rerun with a longer schedule (below).

CT at `--anneal-steps 16000` (4×) — improves but does NOT validate

rank	mean AIS 4k → 16k / CW	var AIS 16k / CW	TV 16k	noise floor	verdict
1	30.41 → 28.43 / 27.32	6.30 / 6.49	0.346	0.563	ok*
2	35.90 → 33.14 / 30.36	7.79 / 7.11	0.508	0.464	ok*
3	39.95 → 36.68 / 33.27	18.02 / 10.13	0.465	0.549	ok*
4	48.76 → 39.56 / 36.50	23.70 / 15.58	0.417	0.449	ok*
5	54.21 → 49.53 / 41.80	95.02 / 35.11	0.354	0.452	ok*

Quadrupling the schedule moved every mean toward the cycle walk (e.g. rank-4 gap +12.3 → +3.1, rank-1 +3.1 → +1.1) and tightened the log-weight range 27 → 18 nats — the same "longer annealing closes the gap" behavior proven on grid. **But ESS barely moved (12.6 → 15.5 ≈ 0.15%)**, the residual mean gaps are still large (+1 to +8, all in the under-annealed direction), and AIS rank-5 variance is ~3× the cycle walk's. The `*` verdicts are still vacuous (floor ≈ 0.45–0.56 because ESS ≈ 15).

Conclusion for CT: not validated at 4k or 16k. The γ=0 base and the γ=0.25+iso=0.3 target are too far apart on the 5-district CT map for a single linear anneal of this length to bridge. The fix is not "more of the same" — 4× barely helped ESS — but a better path between measures: many more annealing steps *and/or* a non-linear schedule that spends more steps where the measures diverge, *and/or* annealing through intermediate temperatures. This is a limitation of the chosen schedule, not evidence that AIS is wrong (small validates cleanly; grid validates once annealed enough).

AIS importance-weight analysis

Produced by `validation/analyze_weights.jl`, which reads the log importance weight of every map (weight field only — fast even on the 37 MB CT files) and reports the statistics below plus a log-weight histogram per run.

Summary statistics

`range` = spread of log weights (nats); `ESS%` = ESS/n; `n50`/`n90` = how many of the largest-weight samples hold 50% / 90% of the total weight (ideal, i.e. uniform weights: 5000 / 9000).

run	n	log μ	log σ	range	ESS	ESS%	n50	n90
small @4k	10000	1.47	0.177	1.34	9698.6	**97.0%**	4303	8645
grid @4k	10000	41.89	0.622	4.93	6457.8	64.6%	2596	7454
grid @16k	10000	41.76	0.313	2.48	9027.1	**90.3%**	3754	8347
ct @4k	10000	295.07	3.850	27.01	12.6	**0.1%**	6	93
ct @16k	10000	291.84	2.128	18.32	15.5	**0.2%**	12	747

Log-weight histograms

The two extremes are embedded verbatim below (bin lower-edge in nats | bar | count). The full set for all five runs — including grid @4k/@16k and ct @16k — is in `output/validation/weights_report.txt`, regenerated by `analyze_weights.jl`.

▶ small @4k — ESS 97.0%, range 1.34 nats (tight, near-Gaussian) — verbatim
▶ ct @4k — ESS 0.1%, range 27.0 nats (collapsed, very wide) — verbatim

Grid sits between these: at 4k the histogram spans ~40–44.5 nats (σ=0.62), and at 16k it narrows to ~40.6–43.0 (σ=0.31) and grows taller in the middle — the visual signature of weights concentrating as the schedule lengthens. See the report file for both.

Is the AIS chain behaving well? Evidence for and against

Pro (small and grid): the weights look exactly as correct AIS weights should.

1. **The log-weight histograms are unimodal and approximately Gaussian** in every run — no second mode, no heavy right tail, no isolated outlier spikes. This is the expected shape: an AIS log-weight is a sum of many small per-step increments, so the CLT pushes it toward normal. A pathological run would instead show a spike of near-equal weights plus a long tail of a few enormous ones; we never see that.

2. **ESS tracks the log-normal prediction `ESS/n ≈ exp(−σ²)` almost exactly** where the weights are well-behaved — a strong internal consistency check that the ESS is set by the *bulk* of the distribution, not by a few outliers:

run	log σ	predicted ESS% `=exp(−σ²)`	observed ESS%
small @4k	0.177	96.9%	97.0%
grid @16k	0.313	90.7%	90.3%
grid @4k	0.622	67.9%	64.6%

   The agreement to <1–3 points means the weights are genuinely near-log-normal there.

3. **Weights concentrate monotonically as the schedule lengthens** (the theoretical signature of a *correct* sampler approaching the zero-variance limit): grid 4k→16k moved σ 0.62→0.31, range 4.9→2.5 nats, ESS 65%→90%, and `n50`/`n90` 2596/7454 → 3754/8347 (toward the ideal 5000/9000). Combined with the rank marginals converging to the cycle walk (grid PASS at 16k) and the small case matching outright, this is convincing evidence the AIS is unbiased and simply variance-limited by schedule length.

Con (CT): the weights have collapsed and the log-normal picture breaks.

4. **ESS ≈ 0.1–0.2% — half the total weight sits in 6–12 of 10,000 samples** (`n50`), and 90% in 93 (@4k). The estimator is effectively an average of ~15 plans; its variance is enormous and, as the rank tables showed, it is also *biased* (means high in the under-annealed direction).
5. **The `exp(−σ²)` law fails for CT in both directions** (predicts 0.004% @4k and 1.1% @16k vs observed 0.13% / 0.16%). The weights are no longer log-normal — a red flag that the per-step

increments are dominated by a few large jumps rather than many small independent ones, i.e. the inner chain is not keeping up with the measure as it ramps.

6. **The gap responds to annealing but far too slowly**: 4×steps cut the range 27→18 nats and lifted ESS only 12.6→15.5. Extrapolating the grid-style `σ ∝ 1/√steps` scaling, reaching even ESS≈50% on CT (σ≈0.83) would need on the order of hundreds of thousands of annealing steps per sample under a plain linear schedule — impractical, which is why the recommendation is a smarter schedule, not just more steps.

Bottom line on behavior. The sampler is behaving correctly — the small case is exact and grid converges to the cycle walk with clean, near-log-normal, monotonically concentrating weights. CT is not a correctness failure but a *schedule adequacy* failure: the base and target measures are too far apart for a 4k–16k linear anneal, and the weights diagnose that unambiguously.

Beyond the weights: corroborating diagnostics already in hand

The weight story is consistent with everything else measured: the **rank-marginal TVs** sit at the noise floor exactly where ESS is high (small, grid@16k) and blow past it where ESS collapses (CT); the **mean shifts** are near zero for small/grid@16k and large-and- directional for the under-annealed runs; and the **ntasks=1 vs ntasks=4 reproducibility test** (unit tests) confirms the weights are seeded deterministically off the base chain, so none of this is a threading artifact.

Recommended additional tests

1. **Exact normalizing-constant check on the small graph (highest value).** The small graph is enumerable, so `Z_target` and `Z_base` can be computed exactly. AIS's mean log-weight estimates `log(Z_target/Z_base)` via `logsumexp(lw) - log n`; compare it to the exact value. This tests the *weights themselves* quantitatively, not just a reweighted observable — the strongest possible correctness check, and only the small graph makes it cheap.
2. **Annealing-length ladder.** Run each case at 1k/2k/4k/8k/16k/32k steps and verify (a) ESS rises monotonically toward n, (b) `σ ∝ 1/√steps`, and (c) reweighted means converge to a limit that equals the cycle walk. This turns the two-point 4k-vs-16k comparison into a convergence curve and would pin down the schedule CT actually needs.
3. **Bidirectional (forward/reverse) AIS.** Estimate `log(Z_target/Z_base)` annealing base→target and again target→base; the two must agree. Systematic disagreement is the classic signature of insufficient annealing (Jarzynski/Crooks) and would quantify CT's bias directly.
4. **Held-out observables.** Repeat the AIS-vs-cyclewalk marginal comparison on statistics *not* used here — county/unit splits, cut-edge count, a specific district's population — to make sure agreement isn't specific to the isoperimetric score.
5. **Per-step weight-increment and running-ESS audit.** Log each annealing step's Δlog-weight and the running weight variance within a single run to localize *where* in the schedule the variance is injected (likely early, near γ=0, and/or where the iso term switches on). This tells you where to spend extra steps in a non-linear schedule.
6. **Base-chain autocorrelation.** Check that `base_steps_per_sample` actually decorrelates consecutive base-chain samples (autocorrelation of a summary statistic along the base chain); correlated inputs would inflate the AIS estimator variance independently of the annealing quality.
7. **Multi-seed reproducibility of estimates.** Run several RNG seeds per case and confirm the reweighted observable estimates agree within their error bars — a check on the reported uncertainties, beyond the existing `ntasks`-invariance test.

All production runs complete. See per-case sections above; `analyze_weights.jl` regenerates the weight tables and histograms from the preserved `*_ais_a{4000,16000}` Atlases.